

Ayush Patel

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EDUCATION

Indian Institute Of Technology Kanpur
Bachelor of Technology (B.Tech) in Chemical Engineering

Nov 2020 - June 2024
CGPA: 7.2/10.0

EXPERIENCE

Tredence Inc. | Data Scientist - Machine Learning, OpenAI, FastAPI Sept 2024 – Present

- Developed KPI-driven test strategies using **K-Means** on image data, improving efficiency and feedback loop by **30%**.
- Enhanced drift detection by **25%** and reduced retraining time by **40%** through a custom drift categorization framework.
- Automated feature pipelines and image labeling using **GPT-4o-mini LLM**, achieving **100%** automation, eliminating manual effort.
- Built **FastAPI**-based real-time dashboards to monitor data reliability, model performance, and trigger retraining.
- Integrated **APM** services into two repos, enabling detailed function-level monitoring and improving visibility in Elastic.

PROJECTS

AI Invoice Auditor - Python, Gemini API, LangChain, LangGraph, Agentic AI, HuggingFace

- Built a **RAG**-based invoice extractor using **LangChain**, **FAISS**, **HuggingFace** embeddings, and **Gemini LLMs**
- Designed a **LangGraph**-based agent with iterative extract-evaluate-optimize feedback loops to self-correct field predictions
- Implemented an **LLM-As-A-Judge** feedback module with Gemini and **Pydantic** to generate field-wise correction signals
- Developed a **Streamlit** web app for uploading invoices and field templates, returning validated structured Excel outputs
- Engineered an evaluation-driven RAG pipeline with feedback prompting and threshold stopping to reduce hallucinations

EMI Default Predictor - Python, Machine learning, Classification

- Developed and evaluated multiple **classification models** to estimate EMI default risk on a highly imbalanced dataset.
- Processed the dataset by applying **one-hot encoding** to categorical variables and **normalizing** numerical features.
- Applied and evaluated classification algorithms such as **Logistic Regression**, **XGBoost**, and **CatBoost** on the dataset.
- Reported model performance metrics, achieving an **Precision of 0.75** and an **Recall of 0.51** with the CatBoost model.

Sign Language Detection - Python, OpenCV, Deep Learning

- Developed a machine learning model to detect **3 types of sign languages** using a **30-frame- sequence** of images as input.
- Generated and preprocessed** the dataset by recording video sample for each sign using the **OpenCV** library.
- Extracted key-point** values from each frame using the **MediaPipe** Holistic library and trained using **LSTM model**.
- Achieved **86% accuracy** on the test dataset and deployed the model for real-time sign language message detection.

SKILLS

Quantitative Analytics: Python, C++, SQL, Data Segmentation

Advanced Analytics & GenAI: AI/ML, Deep Learning, Generative AI, LangChain, LangGraph, HuggingFace, Agentic AI

Dev Tools & Platforms: FastAPI, Azure, VS Code, Github, Git, Postman

ML Frameworks & Libraries: NumPy, Matplotlib, Pandas, PyTorch, TensorFlow, Seaborn, Scikit-learn, OpenCV

Collaboration: Cross-functional Teamwork, Communication, Stakeholder Engagement, Agile Team Collaboration

ACHIEVEMENTS

- Attained All India Rank 3582 in JEE Advanced 2020, ranking among the top 0.25 million shortlisted candidates.
- Implemented key workflows, saving 100+ manual annotation hours and increasing model retraining efficiency by 25%.
- Recognized with a client appreciation award for delivering exceptional results demonstrating strong ownership.
- Finalist in the college-level pitch deck competition Ideathon 2020, conducted by the Alumni Association of IIT Kanpur.
- Organized Techkriti and Udghosh, IIT Kanpur's premier technical and sports festivals, managing a team of 50+ members and coordinating events with 1000+ participants.